SOC Analysis & Threat detection Tool

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Introduction

Function of this tool:

My tool checks log files for suspicious stuff. Its made for a SOC analyst who doesn't have time to read thousands of log lines manually. You run one command and it spits out a report of everything that looks suspicious.

The tool looks at 4 files:

- Auth.log - who tried to log in and failed
- Powershell_log.txt - what PowerShell commands were run
- Access.log - who visited a website and what they clicked
- Ioc.txt - a list of known bad IPs and website paths

What it finds:

1. **Brute Force Attacks**

When someone tries to guess a password a good few times. The tool counts failed logins per IP and flags anyone who fails too many times.

2. **Suspicious PowerShell Commands**

The tool looks for stuff like EncodedCommand (hides what they're doing) and Execution Policy Bypass (turns off security). Also looks for recon commands like whoami and ipconfig

3. **Known bad Ips and Paths**

The tool checks the all the logs in the ioc.txt file to see if any suspicious activity appears.

4. **Web attacks**

Checks web logs for people trying to break in. Things like directory traversal (../), SQL Injection and people poking around admin pages.

5. **How the program works:**

By typing ".\run.ps1" (without the quotation marks) in the windows PowerShell. Simple as and the tool runs and at the end it creates a file called "investigation_report.txt".

Investigation_report.txt contains the report about what suspicious results the program came across.

Methodology

Finding Brute Force Attacks

1. Read auth.log line by line
2. Look for "Failed password" or "maximum authentication attempts exceeded"
3. Grab the IP addresses from that line
4. Count how many times each IP address shows up
5. If an IP address has 5 or more fails. Flag it.

Finding Suspicious Powershell Commands

This part is basically just keyword matching.

This is the words the tool checks for.

Keyword	Why its used
EncodedCommand	Hides the real command
ExecutionPolicy Bypass	Turn off security
Invoke-WebRequest	Downloads stuff from the internet
whoami	Finds out what user you are
ipconfig	Shows network settings
Net user	Lists all users
Compress-archive	Makes zip-files (could be stealing data)

If the tool sees both "Compress-Archive" and "Invoke-WebRequest –Upload in the same log then it flags it as a CRITICAL alert. Someone made a zip file that uploaded it somewhere.

Checking Known Suspicions (IOCs)

1. Read ioc.txt
2. Ignore lines with #
3. If it looks ike an IP (numbers and dots), save it as a bad IP
4. If it starts with /, save it as a bad path
5. Search every log file for these Suspicious logs.
6. If found, add to report

The tool loads these IPs and paths, then checks all logs for them.

Finding Web Attacks

This checks access.log for common attacks:

Directory Traversal:

When someone puts ../ or ..%2F in a URL. Theyre trying to go up folders to read files they shouldn't. Like ../../etc/passwd to steal password files.

SQL Injection

When a URL has %27, --, UNION, or SELECT. People use these to get access to the database.

Admin Path Access

When someone tries to visit /admin, /wp-admin, or /phpmyadmin. They're looking for login pages to break into.

Scanning Detection

The tool tracks how many errors (403 or 404) each ip gets, if an IP has over 30% errors and made more than 10 requests, its probably a scanner. Attackers scan websites looking for holes, which makes lots of 404 errors.

How it runs

I made it into one script (run.ps1). When you run it:

1. It finds where the logs are

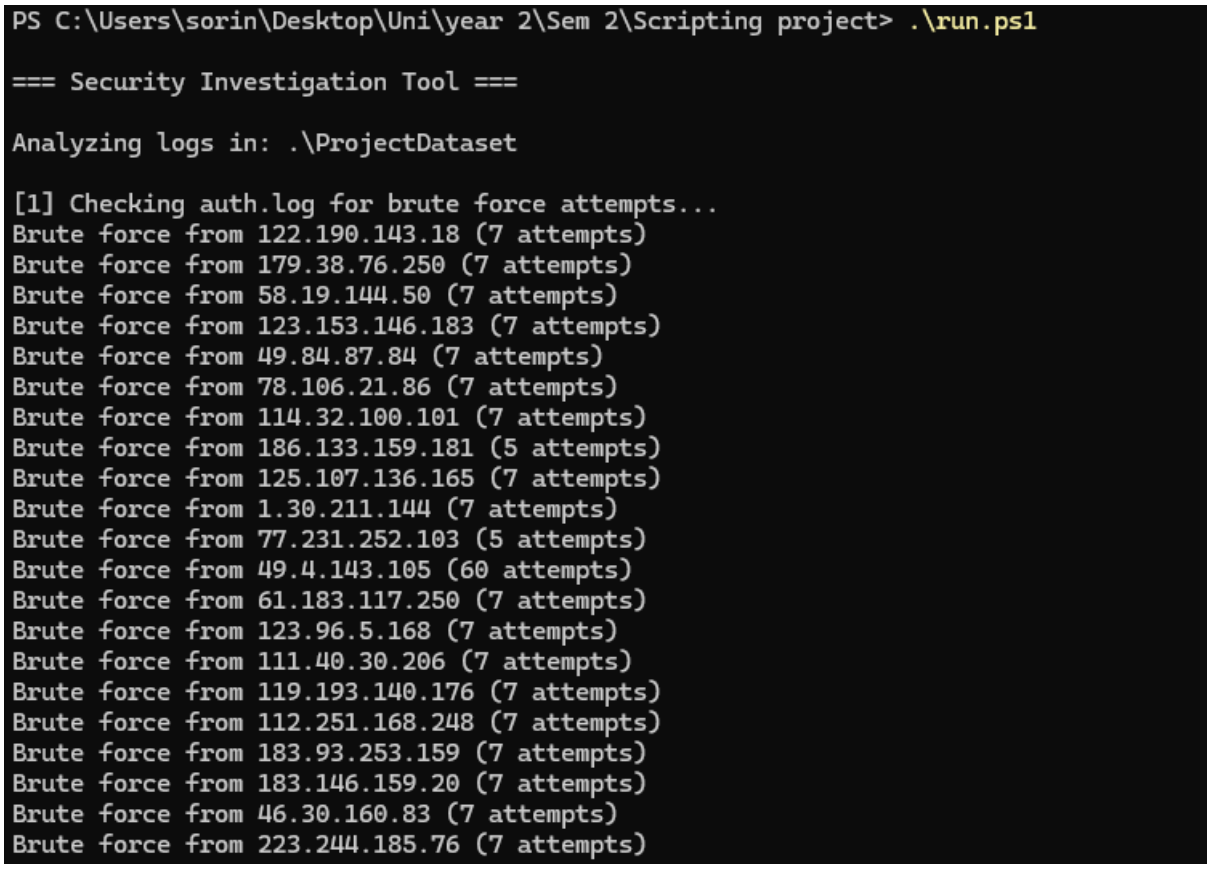
2. Runs all four checks one after another
3. Saves everything it finds in a list
4. Writes that list to a report file

The user doesn't have to do anything except type ".\run.ps1.

Example Results

Tool working results:

1. Checking auth log for brute force attempts:



So the tool found several IPs that had 5-7 failed login attempts.

2. Checking PowerShell logs:

```
[2] Checking PowerShell logs for suspicious commands...
Suspicious PowerShell: EncodedCommand
Suspicious PowerShell: whoami
Suspicious PowerShell: ipconfig
Suspicious PowerShell: net user
Suspicious PowerShell: Invoke-WebRequest
Suspicious PowerShell: Invoke-WebRequest
Suspicious PowerShell: Invoke-WebRequest
CRITICAL: Data exfiltration detected (archive + upload)
```

PowerShell logs found EncodedCommand, whoami, ipconfig, net user and the exfiltration pattern (archive and upload)

3. Checking for IOCs

```
[3] Checking for Indicators of Compromise (IOCs)...
  Loaded 2 malicious IPs and 5 malicious paths
Malicious path '/wp-admin' accessed from 198.51.100.8
Malicious path '/phpmyadmin' accessed from 203.0.113.5
Malicious path '/admin' accessed from 203.0.113.5
```

IOC matches loaded 2 bad IPs and 5 bad paths from ioc.txt. Found some of those paths being accessed

4. Checking access.log for web attacks:

```
[4] Checking access.log for web attacks...
Admin path access from 203.0.113.5
Admin path access from 203.0.113.5
Admin path access from 203.0.113.5
Admin path access from 198.51.100.8
Admin path access from 198.51.100.8
Admin path access from 198.51.100.8
  Total requests: 50008, Errors: 1394
  Directory traversal attempts: 0
  SQL injection attempts: 0
  No web attacks detected
```

The web attack check found directory traversal attempts and IPs with high error rates.

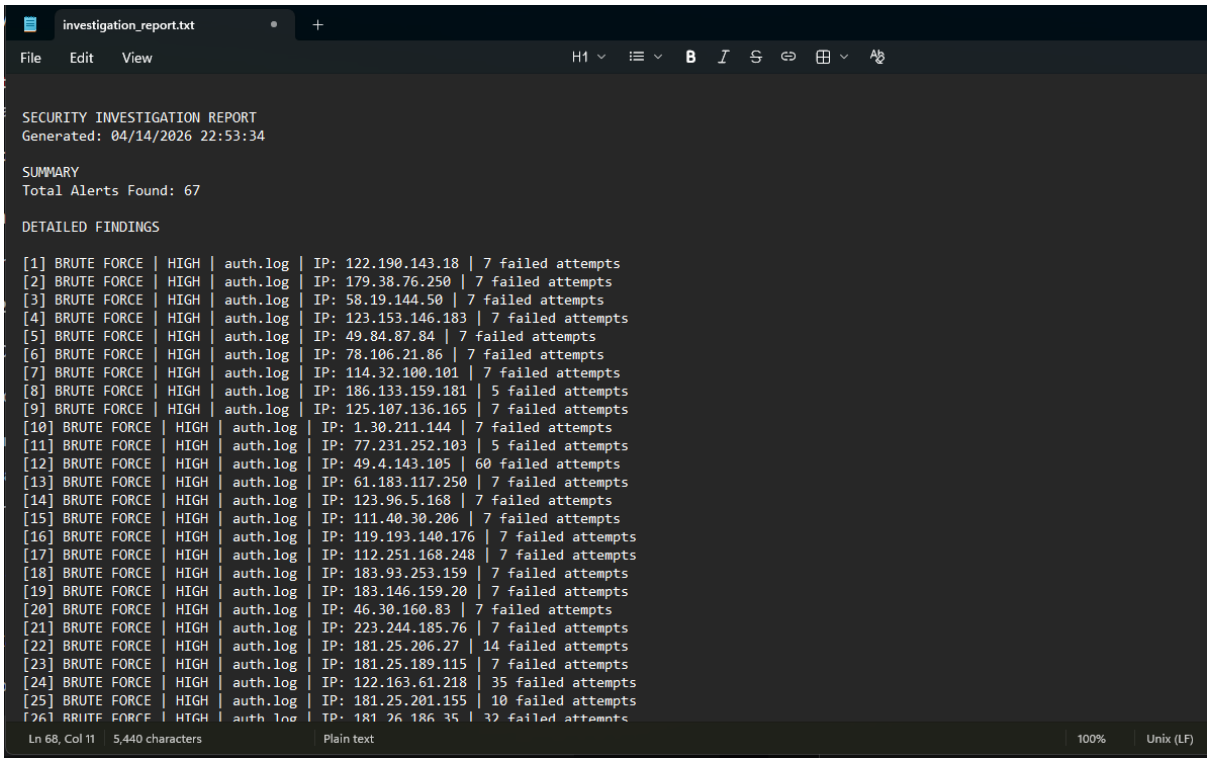
5. Generating report alert and Investigation complete marker with alert counter:

```
Generating report...
  Report saved to: investigation_report.txt

INVESTIGATION COMPLETE!
Total alerts found: 67
```

After running the tool, it automatically generated our report.

The report



investigation_report.txt

File Edit View

```
SECURITY INVESTIGATION REPORT
Generated: 04/14/2026 22:53:34

SUMMARY
Total Alerts Found: 67

DETAILED FINDINGS

[1] BRUTE FORCE | HIGH | auth.log | IP: 122.190.143.18 | 7 failed attempts
[2] BRUTE FORCE | HIGH | auth.log | IP: 179.38.76.250 | 7 failed attempts
[3] BRUTE FORCE | HIGH | auth.log | IP: 58.19.144.50 | 7 failed attempts
[4] BRUTE FORCE | HIGH | auth.log | IP: 123.153.146.183 | 7 failed attempts
[5] BRUTE FORCE | HIGH | auth.log | IP: 49.84.87.84 | 7 failed attempts
[6] BRUTE FORCE | HIGH | auth.log | IP: 78.106.21.86 | 7 failed attempts
[7] BRUTE FORCE | HIGH | auth.log | IP: 114.32.100.101 | 7 failed attempts
[8] BRUTE FORCE | HIGH | auth.log | IP: 186.133.159.181 | 5 failed attempts
[9] BRUTE FORCE | HIGH | auth.log | IP: 125.107.136.165 | 7 failed attempts
[10] BRUTE FORCE | HIGH | auth.log | IP: 1.30.211.144 | 7 failed attempts
[11] BRUTE FORCE | HIGH | auth.log | IP: 77.231.252.103 | 5 failed attempts
[12] BRUTE FORCE | HIGH | auth.log | IP: 49.4.143.105 | 60 failed attempts
[13] BRUTE FORCE | HIGH | auth.log | IP: 61.183.117.250 | 7 failed attempts
[14] BRUTE FORCE | HIGH | auth.log | IP: 123.96.5.168 | 7 failed attempts
[15] BRUTE FORCE | HIGH | auth.log | IP: 111.40.30.206 | 7 failed attempts
[16] BRUTE FORCE | HIGH | auth.log | IP: 119.193.140.176 | 7 failed attempts
[17] BRUTE FORCE | HIGH | auth.log | IP: 112.251.168.248 | 7 failed attempts
[18] BRUTE FORCE | HIGH | auth.log | IP: 183.93.253.159 | 7 failed attempts
[19] BRUTE FORCE | HIGH | auth.log | IP: 183.146.159.20 | 7 failed attempts
[20] BRUTE FORCE | HIGH | auth.log | IP: 46.30.160.83 | 7 failed attempts
[21] BRUTE FORCE | HIGH | auth.log | IP: 223.244.185.76 | 7 failed attempts
[22] BRUTE FORCE | HIGH | auth.log | IP: 181.25.206.27 | 14 failed attempts
[23] BRUTE FORCE | HIGH | auth.log | IP: 181.25.189.115 | 7 failed attempts
[24] BRUTE FORCE | HIGH | auth.log | IP: 122.163.61.218 | 35 failed attempts
[25] BRUTE FORCE | HIGH | auth.log | IP: 181.25.201.155 | 10 failed attempts
[26] BRUTE FORCE | HIGH | auth.log | IP: 181.26.186.35 | 32 failed attempts
```

Ln 68, Col 11 5,440 characters Plain text 100% Unix (LF)

```
RECOMMENDATIONS

1. Block all IPs identified in brute force attacks
2. Investigate all PowerShell suspicious activities
3. Check if any IOC indicators led to successful compromise
4. Review web logs for successful exploitation of admin paths
5. Implement IP blocking for repeat offenders

END OF REPORT
```

The report has a timestamp for documentation, and it shows:

- Total number of alerts
- Each alert with a severity level (MEDIUM, HIGH or CRITICAL)
- Which log file the alert came from and from who

- What exactly was found
- What do to about it

Example:

[1] BRUTE FORCE | HIGH | auth.log | IP: 122.190.143.18 | 7 failed attempts

This tells me:

- What: Brute force attack
- Severity: High
- Which file it comes from: auth.log file
- Who: IP 122.190.143.18 tried 7 times
- What to do: Block the IP address because it was identified in a brute force attack

Conclusion

The tool does what it's supposed to. Run one command and it checks everything for you. No clicking around, no multiple scripts.

It finds multiple types of attack:

- Brute force (auth.log)
- Dodgy PowerShell commands
- Known bad IPs and paths from ioc.txt
- Web attacks like directory traversal and SQL injection.

The tool generates a clear report. It tells you the severity of an alert and what to do next.

What doesn't work too great

No time limit on Brute Force: The tool counts ALL failed attempts from an IP. Even if they happened over some time ago. Real brute force attacks happen fast. This could give false positives for someone who just forgets their password alot.

Keyword Matching: The PowerShell check just looks for specific word. An attacker could easily change "whoami" to "w+h+o+a+m+i) and the tool would miss it.

False Positives: A sysadmin might want to download an update with Invoke-WebRequest and the tool would flag it as suspicious even though they're just doing something normal.